



Educational Challenges of the Digital World

MedTech Education, Train-the-Trainer
and Professional Networking
Conference

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Foreword

This volume presents abstracts from the Educational Challenges of the Digital World - MedTech Education, Train-the-Trainer and Professional Networking Conference, held in Budapest on 7 September 2026 as part of the Wekerle Conference Series. As a session chair, what I remember most is the diversity of the presentations and the thought-provoking questions and discussions. The contributions collected here reflect this diversity, offering different perspectives on a shared challenge: how artificial intelligence and digital technologies are changing education and our lives.

Reviewing the volume brought to mind Cixin Liu's *The Three-Body Problem*. When the fundamental rules of reality change, previous ways of thinking may no longer be good enough. Artificial intelligence presents us with a similar intellectual challenge. We are trying to understand and keep pace with rapidly evolving technology while also trying to grasp how it changes the knowledge and skills we teach, and even our assumptions about teaching and learning. It is perhaps no surprise that conversations about artificial intelligence so naturally become conversations about the future.

The volume follows the conference's thematic structure. It begins with a timely question: what does artificial intelligence mean for the way we teach and learn? The opening contributions examine AI and digitalization in education from different perspectives, including organizational readiness, ethics, personalized learning, and student motivation. Together, they highlight the complex relationship between technology, pedagogy, educational institutions, and learners.

The focus then shifts towards safety and security. Contributions on compliance, cybersecurity, cloud-based learning environments, and the responsible use of AI remind us that innovation cannot be separated from trust, regulation, and responsibility. New technologies create new opportunities, but also new risks and dependencies.

The next group of contributions focus on MedTech and healthcare education, where the connection between technological innovation and education becomes particularly tangible. Topics ranging from cybersecurity and AI-enabled medical technologies to public health education, living labs, and virtual PK-PD digital twins demonstrate the growing need to combine technological, clinical, regulatory, and professional knowledge.

Several contributions take a practical perspective, asking a fundamental question: can new technologies truly help learners acquire, understand, and apply knowledge more effectively? The final papers return to educational methodology, the learner, and the role of reflection in the learning process.

The abstracts in this volume reflect a wide range of professional backgrounds and approaches. I hope readers will find them as thought-provoking as I found the presentations and that they will inspire further questions and new ideas.

Dr. Petra Benedek
Executive Vice-Rector
Wekerle International University

Dr. Aisi Li

(Wekerle International University)

Learning to Transform: An Educational Perspective on Organisational Readiness for AI and Digitalisation;

Digitalisation and artificial intelligence are transforming education, training and professional practice faster than many organisations can adapt. While technologies and stakeholder expectations change rapidly, institutional strategies, staff competencies, governance arrangements and quality-assurance processes often require considerably more time to develop.

This presentation examines the resulting gap between stakeholder expectations and organisational readiness from the perspective of educational studies, higher education leadership and international cooperation. It argues that digital readiness is not simply access to technology, but an organisation's capacity to use it purposefully, responsibly and sustainably. Effective transformation requires alignment among strategic objectives, staff competencies, working processes, governance, institutional culture, financial capacity and evaluation. It also depends on learning at three interconnected levels: individual, organisational and system-wide. Particular attention is given to Train-the-Trainer initiatives as a means of building sustainable organisational capacity. Effective trainers do more than transfer information: they adapt knowledge to different contexts, facilitate workplace application, encourage reflective practice and help embed new competencies in organisational routines. This requires institutional support, opportunities for practice and conditions that enable employees to apply their learning.

The presentation proposes a practical readiness cycle based on listening to stakeholders, assessing institutional capacity, prioritising strategically relevant changes, building human capability, and governing and evaluating implementation. It concludes that although digital transformation may begin with technology, its success ultimately depends on the capacity of individuals, organisations and professional networks to learn and adapt continuously.

Elizaveta Glazova

(Higher School of Economics)

AI Ethics in Education: Main Challenges of Teaching, Learning and Assessment with Artificial Intelligence

Artificial intelligence is becoming part of the educational process through applications that support lesson planning, content creation, translation, tutoring, feedback, assessment preparation, and administrative work. These applications create significant opportunities, but they also raise ethical questions about privacy, bias, transparency, academic integrity, unequal access, accuracy, human autonomy, and the changing responsibilities of teachers and learners. This paper examines the main ethical challenges associated with the use of artificial intelligence in education and proposes a practical framework for responsible implementation.

The paper uses a conceptual and practice-oriented approach. It analyses the educational process as a sequence of connected activities — planning, teaching, learning, assessment, and feedback — and identifies the ethical risks that may arise at each stage. The proposed framework organises these risks into five areas: data and privacy; fairness and inclusion; transparency and accountability; human agency and professional judgement; and integrity and quality of learning. For each area, the paper identifies questions that educators and institutions should address before introducing an AI-supported activity.

The paper argues that responsible AI use requires more than technical competence or a general institutional policy. Teachers and learners need context-specific AI literacy, clear rules for acceptable use, methods for verifying AI-generated information, mechanisms for disclosing AI assistance, and assessment practices that preserve meaningful evidence of individual learning. The article concludes that ethical AI integration should be understood as an ongoing educational and organisational process rather than a one-time compliance exercise. The framework is intended to support educators, trainers, and institutions in adopting useful AI applications while protecting human dignity, educational quality, privacy, fairness, and professional responsibility.

Rozália Szatmáry

(Obuda University)

Generative AI as Individualized Learning Support: Opportunities, Limits, and Pedagogical Effectiveness

Learners within the same educational setting often differ considerably in prior knowledge, learning pace, confidence, educational background, and preferred forms of support. As students progress through education, these differences may become increasingly pronounced, particularly in interdisciplinary and professionally oriented programmes. Although individualized support is widely regarded as educationally valuable, limited classroom time, large student groups, and differences in learner needs may restrict the extent to which teachers and trainers can respond to every student individually.

Supplementary learning resources have therefore become increasingly important. Students may rely on private tutoring, online platforms, educational videos, digital repositories, and, more recently, generative artificial intelligence. Compared with many traditional digital resources, generative AI offers immediate, interactive, and apparently individualized support. Learners can ask specific questions, request alternative explanations, seek clarification, and continue the interaction until a response appears understandable.

However, accessibility and responsiveness should not automatically be equated with educational effectiveness. The ability of an AI system to provide an answer does not necessarily indicate that meaningful learning has occurred. Educational usefulness must also be considered in relation to the accuracy of the information provided, the quality of conceptual explanation, adaptation to the learner's level of prior knowledge, recognition of misconceptions, appropriate scaffolding, feedback, and alignment with intended learning outcomes.

The potential and limitations of generative AI as an individualized learning support tool are therefore examined through a critical review of the scientific literature. A distinction is made between task performance and pedagogical effectiveness. Particular attention is given to whether generative AI supports understanding, transfer, learner independence, and sustained knowledge development, rather than merely enabling the rapid completion of individual tasks.

A broader framework is proposed for evaluating AI-supported learning tools from an educational rather than solely technological perspective. Such evaluation should consider not only whether an answer is correct or immediately available, but also whether the interaction supports the cognitive processes required for learning. Generative AI may provide substantial opportunities for more accessible and flexible learning support, but its educational value ultimately depends on how effectively availability of information is transformed into understanding.

Rodion Rakhmatullin

(Saint Petersburg Electrotechnical University; ITMO University)

Integrating Artificial Intelligence into Innovation Management Education: An Analysis of the Bachelor's Programme at ETU "LETI"

The rapid development of artificial intelligence (AI) is transforming not only business processes but also the competencies required from future innovation and technology management professionals. This study examines how AI is being integrated into the Bachelor's programme in Innovation Management at Saint Petersburg Electrotechnical University "LETI".

The research is based on a comparative analysis of the curricula and course descriptions within the educational programme "Information Systems and Technologies in Innovation Activities". Particular attention is paid to the transformation of courses related to marketing and human resource management. The analysis demonstrates a shift from predominantly traditional approaches towards the systematic integration of AI-based tools, analytics and automation into professional disciplines.

For example, the updated course "Digital Marketing Technologies" explicitly incorporates AI into market analysis, consumer behaviour analysis, marketing strategy development, content generation, communication automation and decision-making. Similarly, the new course "Digital HR Analytics" integrates AI and machine learning into recruitment, employee development, performance assessment and predictive HR analytics. These changes indicate that AI is increasingly treated not as a standalone technological subject, but as a cross-functional competence relevant to innovation management.

The study proposes a framework for analysing AI integration into higher-education curricula based on three dimensions: transformation of course content, development of practical AI-related skills, and evolution of learning outcomes and professional competencies.

The findings highlight the importance of continuously adapting innovation management curricula to technological change. The experience of ETU "LETI" demonstrates how higher-education programmes can combine classical management and innovation theories with emerging AI technologies while maintaining a strong focus on practical and interdisciplinary competencies.

The study concludes with recommendations for further development of AI-related educational content and competencies in innovation management programmes.

Dr. Ádám Béla Horváth

(Obuda University)

Why Learners Adopt and Benefit from MOOCs: A Systematic Review of Drivers, Barriers, Motivation, and Realised Benefits

Massive open online courses (MOOCs) have become an important component of flexible higher education and continuing professional development. Their scalable and accessible format is particularly relevant to rapidly changing interdisciplinary fields such as medical technology, where professionals must regularly update technical, clinical, and regulatory knowledge. Nevertheless, enrolment does not necessarily result in sustained participation, course completion, or workplace-relevant learning, while completion rates alone may underestimate success when learners intentionally study only selected modules.

This systematic review examines four interconnected questions: which factors drive learners to adopt MOOCs; which barriers restrict participation and persistence; what motivates continued engagement; and which educational, professional, and personal benefits learners actually realise. Following PRISMA 2020, Scopus, Web of Science, ERIC, and ScienceDirect will be searched for English-language, peer-reviewed empirical studies published between 2015 and August 2026. Eligible studies will address adult, higher-education, or professional learners and report evidence concerning at least one focal domain. The findings will be synthesised using framework-based thematic analysis and descriptive evidence mapping.

A preliminary assessment of fifteen foundational and recent studies indicates that perceived usefulness, flexibility, accessibility, course relevance, reputation, and performance-to-cost value are prominent adoption drivers. Motivation, self-regulation, satisfaction, and engagement appear to mediate persistence and benefit realisation. Major barriers include limited time, competing obligations, insufficient prior knowledge or digital skills, weak interaction and feedback, unsuitable course design, language and infrastructure constraints, and inadequate recognition of credentials. Realised benefits extend beyond certificates to knowledge acquisition, professional development, workplace application, personal growth, and increased learning autonomy.

The review will integrate determinants frequently examined in isolation and translate them into practical design principles. For MedTech education, these findings support modular pathways, explicit workplace relevance, differentiated learner scaffolding, credible competency assessment, expert interaction, and recognition mechanisms suitable for heterogeneous professional audiences. These principles may also strengthen blended Train-the-Trainer programmes and institutional cooperation.

Dr. Mayibongwe Tafara Mudzengi

(INTI International University)

Turning Training Investment Into Impact: The Motivation Multiplier

Undoubtedly, training in the education sector plays a crucial role in fostering human development in Malaysia. The Malaysia Education Blueprint (2013–2025) emphasizes the adaptation of education to meet the requirements of the Fourth Industrial Revolution, highlighting a deliberate shift towards developing well-trained and scientifically skilled individuals. This research's primary goal is to model the influence of transfer of training on cost-effectiveness in Private Higher Education in Malaysia: the role of motivation as a mediator. The focus on the cost-effectiveness of training within the Malaysian education sector guided by the Transfer of Training Model by Baldwin and Ford (1988) and the Kirkpatrick Model by Donald Kirkpatrick (1950) will provide an insight into the influence of transfer of training and what measures are essential for cost-effectiveness.

However, training programs are frequently not adequately reviewed to measure investment return. Evaluation of the transfer of training is required for the further realization of the investment return made by stakeholders. Motivation being the mediator can be redefined as comprehensive training motivation. The sample size consisted of Human Resource (HR) personnel from Executives to Vice Presidents in the Malaysia Private Higher Education sector who either initiate or design training for academic staff in institutions' current stat of 29,413 personnel (Ministry of Higher Education Malaysia, 2022). Random sampling technique was utilized, based on the assumption of an average of five (5) human resource personnel in 94 Private Higher Education, comprised of private universities (50), private university colleges (34), and foreign university branch campuses (10). The quantitative method was employed for this study with Cohen's (1992) sampling table determining sample size determined the desired sample size was 256 surveys to be distributed.

Structural Equation Modeling (SEM) is used to simultaneously visualize and validate several statistical associations. The analysis of the study's data revealed that three hypotheses were supported and seven were not supported, indicating that these transfer of training attributes may not be significant predictors of cost-effectiveness in the transfer of training in Private Higher Education.

Prof. Dr. Kornélia Lazányi

(Wekerle International University; TAMCert)

AI and the Promise of Frictionless Working

Artificial intelligence is increasingly associated with the promise of frictionless learning and working. Explanations, summaries, solutions, and polished texts can be generated within seconds, allowing tasks to be completed with less time and apparent effort. Although such efficiency may improve access and reduce unnecessary barriers, it cannot automatically be assumed that the removal of effort produces better learning.

The educational implications of frictionless AI-supported learning are examined through cognitive psychology and Daniel Pink's motivational framework. Research on retrieval practice, the generation effect, desirable difficulties, and productive failure suggests that effort may form part of the learning process itself. Knowledge has been shown to be retained more effectively when it is actively retrieved rather than repeatedly reviewed (Roediger & Karpicke, 2006), while information generated by learners is often remembered more successfully than information that is simply received (Slamecka & Graf, 1978). Under appropriate conditions, initial difficulty, uncertainty, and unsuccessful attempts may prepare learners for deeper understanding and later transfer (Bjork & Bjork, 2011; Kapur, 2008).

When a complete and convincing response is provided by AI before an independent attempt has been made, learning may be shifted from active generation towards passive recognition. Immediate task performance may be improved, while the learner's ability to retrieve, explain, or apply the knowledge independently remains underdeveloped. However, not all difficulty is educationally valuable. A distinction must be made between extraneous friction, which consumes cognitive resources without supporting learning, and productive friction, through which reasoning, retrieval, reflection, and revision are encouraged.

A related motivational concern may be identified through Pink's concepts of autonomy, mastery, and purpose (Pink, 2009). Autonomy may be reduced when decisions and approaches are delegated to AI. Mastery may be weakened when successful output is produced without a corresponding sense of increased personal capability. Purpose and ownership may also be diminished when little intellectual investment has been made in the completed work. These concerns are consistent with self-determination theory, in which autonomy and competence are regarded as important conditions of intrinsic motivation (Ryan & Deci, 2000).

The value of AI in education should therefore not be assessed only through speed, convenience, or output quality. Attention must also be given to the cognitive and motivational processes removed from the learning experience. Unnecessary barriers may appropriately be reduced, but opportunities for retrieval, generation, uncertainty, error, and revision should be deliberately preserved. Effective AI-supported learning may depend not on the elimination of all friction, but on the careful design of the right kind of friction.

Dániel Frankl

(Obuda University)

Interactiv learning aided by immersive technologies

The evolution of Virtual Reality (VR) hardware has fundamentally transformed spatial computing from niche experimental setups into accessible educational tools. Understanding the historical progression of these devices, from cumbersome, tethered systems with low-resolution displays to modern, standalone headsets, provides essential context for their current pedagogical value. Today's immersive technologies offer untethered freedom and precise tracking capabilities, finally providing the technical foundation necessary to support widespread, meaningful interactive learning. Building upon this hardware maturation, this presentation introduces targeted research focusing on the application of VR in highly specialized, hazardous training scenarios, specifically explosive device familiarization and bomb recognition. Traditional training in these high-stakes fields often relies on two-dimensional manuals or inert models that fail to convey spatial complexity. This research demonstrates how immersive technologies bridge that gap by effectively simulating critical, life-threatening situations with absolute physical safety. By allowing trainees to inspect and interact with highly detailed, real-scale virtual models of explosive devices, VR facilitates crucial spatial awareness. Trainees can safely examine intricate mechanisms and practice rapid decision-making, highlighting the immense value of experiential learning where real-world practice is too dangerous, cost-prohibitive or logistically impossible. Expanding beyond these specialized hazard scenarios, the presentation evaluates the broader implications of VR and Augmented Reality (AR) in general education and vocational training. The core advantage demonstrated in the bomb recognition research, active, hands-on participation translates directly to conventional academic disciplines. Immersive technologies are actively shifting the educational paradigm away from passive information consumption toward active, experiential learning. Whether students are practicing complex mechanical repairs or exploring historically accurate digital reconstructions, VR allows them to learn by doing. By seamlessly embedding these interactive tools into everyday school curriculums and training programs, educators can foster dynamic environments that improve student engagement, enhance critical thinking and increase long-term knowledge retention.

Sára Szatmáry

(Obuda University)

*Beyond E-Learning: Matching Learning Outcomes to Digital and Experiential
Formats*

The expansion of e-learning has created new opportunities for accessible, flexible, and scalable education. Knowledge-based content can increasingly be delivered through digital materials, allowing learners to prepare independently, revisit complex topics, and progress at an individual pace. In professional and interdisciplinary fields such as MedTech, these possibilities are particularly valuable, as learners may enter programmes with different educational backgrounds, levels of prior knowledge, and professional experience. However, the increased availability of digital learning also raises an important design question: should all learning outcomes be approached in the same way simply because content can be delivered online?

Not all educational objectives are equally suited to fully digital formats. While theoretical knowledge, preparatory materials, and conceptual foundations may often be learned effectively through e-learning, other outcomes depend more strongly on active participation, interaction, and experience. Communication, teamwork, practical problem-solving, professional judgement, negotiation, adaptation, and intercultural competence are developed not only through exposure to information, but through situations in which knowledge must be applied in interaction with others. From this perspective, the value of blended and short intensive programmes lies not merely in combining online and face-to-face elements, but in deliberately assigning different learning outcomes to the environments in which they can be developed most effectively. Digital components may be used for preparation, knowledge acquisition, individual study, and reflection, while face-to-face activities can be reserved for collaboration, simulation, discussion, peer learning, feedback, and the application of knowledge in complex or socially situated contexts.

Intercultural competence provides a particularly clear example. Digital resources may increase awareness of cultural differences and provide access to international perspectives, but first-hand interaction requires learners to respond to ambiguity, adjust communication, negotiate misunderstandings, and cooperate with people whose assumptions or experiences may differ from their own. Similar principles apply to many professional and interpersonal competences relevant to contemporary education and training.

Effective blended learning should therefore not be understood as the simple division of teaching time between online and physical environments. Its pedagogical potential lies in matching learning outcomes with the format that can support them most effectively. As e-learning becomes more sophisticated, face-to-face learning may become more valuable rather than less, provided that it is used purposefully for engagement, interaction, application, and skill development.

Gerald Korir

(Budapest University of Technology and Economics)

A Process-Oriented Model for Quality and Digital Transformation in Higher Education.

Higher education institutions operate in an increasingly complex environment in which quality assurance, accountability and institutional adaptability must be addressed simultaneously. Digital transformation, changing stakeholder expectations and new forms of educational delivery require institutions not only to demonstrate compliance with quality standards, but also to develop processes that support continuous learning, coordination and improvement. This research examines these challenges through a process-oriented perspective on educational service delivery.

The study develops a conceptual model integrating Business Process Management (BPM), ISO 9001:2015 and the Standards and Guidelines for Quality Assurance in the European Higher Education Area (ESG). The model organizes institutional activities into three interconnected layers: core educational processes, support and enabling processes, and governance and quality-assurance processes. Each layer follows an input–process–output–stakeholder logic and is embedded in the Plan–Do–Check–Act (PDCA) cycle. BPM provides the operational structure through process ownership, documentation and performance measurement, while ISO 9001:2015 and ESG provide complementary principles for systematic quality assurance and continuous improvement.

The research addresses a key institutional challenge: quality assurance can become primarily a compliance exercise when processes, responsibilities, evidence and feedback mechanisms are insufficiently connected. A process-oriented approach instead positions quality assurance as an instrument for institutional learning, coordination and improvement. It also provides a foundation for evaluating and managing digital transformation by connecting technological change with educational and organizational processes.

The study further considers institutional adaptability and the risk of mission drift in higher education. Complementary insights from research on positive psychology in higher education, management and vocational rehabilitation broaden the perspective by examining how institutions and their members respond to changing organizational demands. Taken together, these perspectives highlight the importance of maintaining institutional purpose while adapting processes, practices and capabilities to changing educational and workplace environments. The research therefore positions quality, process maturity, adaptability and institutional mission as interconnected dimensions of sustainable higher-education development.

Dr. Eszter Metz

(Sapientia Hungarian University of Transylvania)

Teaching Data and Network Visualization in Public Health Education: A Ten-Module Course for Building Systems Thinking

Public health problems are rarely isolated: they emerge from systems of actors, institutions, behaviours and information flows. Curricula, however, still largely train students to read isolated indicators, leaving graduates under-prepared for relational reasoning and critical visual interpretation. This contribution presents the design and outcomes of a one-semester course that embeds data visualization and network science into an undergraduate public health programme, with the pedagogical aim of shaping systems thinking rather than transmitting software skills.

The course runs within the Bachelor-level Public Health programme of Sapientia Hungarian University of Transylvania (180 ECTS; cohorts of approximately 20 students). It comprises ten modules: data and the attention economy; the data-analysis model; data collection from Romanian and European databases; data cleaning and preparation; principles of data visualization; network science and network visualization; misinformation and design criteria; graphics, colour and perception; maps and spatial epidemiology; and infographics. Each module pairs a short theoretical input with hands-on analysis of real public health datasets. Students construct and interpret network graphs of services and institutions, apply centrality measures to identify key actors, and use clustering to reveal at-risk communities and inequalities in access. A dedicated strand on data deception trains the critical reading of truncated axes, cherry-picking, spurious correlation and false balance, framed by Bertin's criteria of expressive and effective representation. Assessment is portfolio-based: each student produces a visual and network analysis of a self-selected public health question.

Across cohorts, students shift from indicator-based to relational description: they identify structural vulnerabilities and key actors rather than reporting single values, justify their graphical choices, and detect misleading encodings in published material. Network visualizations function as a shared language across professional backgrounds, supporting interdisciplinary dialogue.

Data and network visualization is not a technical add-on but an essential competency for analytical clarity and systems-oriented reasoning. The ten-module structure is transferable, and datasets and exercises can be shared on request.

Nikita Kalganov

(Obuda University)

LLM-based Agent Predicting Exchange Rate Fluctuations to Advise International Students of Tuition Fee Payment Date

Internationalization of medical education has contributed to cross-border student mobility, encouraging medical students to pursue their studies outside their home countries. International students often face significant financial uncertainty when tuition fees are denominated in foreign currencies, as unexpected exchange rate fluctuations can increase the final cost of study abroad. If international students do not plan tuition payment date in advance, the amount payable for the next semester can increase their financial burden in their domestic currency. Universities can support international students in selecting a more favourable tuition fee payment date by introducing artificial intelligence (AI) agents forecasting short-term currency exchange rate trends. Large language model-based (LLM) AI agents can orchestrate other programs with limited human oversight, collecting data from external sources, executing forecasts, and interpreting the results in a human-readable format. This study proposes the concept of a prototype AI agent designed to collect European Central Bank (ECB) data on exchange rates, evaluate their short-term trends, and advise the student of a potentially more favourable currency exchange date to pay the tuition fee. The research develops and demonstrates an AI agent solution integrating an LLM-based advisory layer with a structured exchange rate data retrieval process and a statistical forecasting component. The agent retrieves daily EUR exchange rate reference data from the ECB Application Programming Interface (API) and stores the observations in a local database for further interpretation. The current prototype focuses on the EUR/HUF currency pair and uses historical ECB data from January 2021 to April 2026 for the initial training. A Linear Regression model is retrained with newly retrieved observations after each user query, documenting Root Mean Square Error (RMSE) to monitor model performance. Based on the student's tuition amount, invoice date, payment deadline, and domestic currency, the prototype system is designed to forecast exchange rates for the subsequent two weeks and identify a potentially preferable currency conversion date. An LLM-based advisory layer presents the forecast and explains the resulting recommendations on tuition fee payment date in student-oriented language. The proposed solution is intended to provide a data-driven approach to tuition payment decisions, with a potential to reduce financial stress from studying abroad, improve international student engagement, and encourage participation in extracurricular activities instead of handling mundane financial problems. The prototype solution demonstrates the technical feasibility of integrating exchange rate retrieval, statistical forecasting, and LLM-based interpretation in a single decision support system. Current solution is limited to one currency pair and a simple forecasting model, providing the proof-of-concept but requiring further evaluation of predictive accuracy and relevance of results. The study contributes to emerging research on AI-assisted student services by illustrating a practical application of AI agents in higher education administration, financial decision support, and international student welfare.

Dr. Ákos Ádám Molnár

(TAMCert)

The Role of Compliance in Building a Safer Future

Presenting and Corresponding Author Rapid technological advancements in artificial intelligence, cloud-based healthcare data management, and software-driven medical devices have fundamentally transformed the digital landscape. However, these innovations have outpaced traditional physical safety paradigms. In the contemporary environment, the concept of safety is no longer limited to physical harm; it has evolved into a multidimensional construct that necessitates the strict integration of cybersecurity, data privacy, and ethical artificial intelligence governance. This presentation aims to redefine the modern concept of safety and demonstrate how rigorous regulatory compliance serves as the essential mechanism for building societal trust and ensuring the market viability of disruptive digital technologies. The work conducts a structural analysis of the relationship between mandatory legislative frameworks and technical engineering standards. It utilizes a comparative methodology to distinguish between traditional physical risk assessment, focusing on health hazards and physical safety and modern impact assessment, focusing on fundamental human rights, algorithmic bias, and societal fairness. The conceptual framework explores how harmonized standards act as a practical bridge, translating legal requirements into actionable technical solutions. The analysis reveals that isolated technological performance, such as high algorithmic accuracy in clinical diagnostics, is insufficient for market adoption without corresponding privacy and cybersecurity measures. The research establishes a three-tiered "Golden Rule" for modern regulation, regulations define societal goals, technical standards provide the operational framework, and independent conformity assessment provides the vital evidence of compliance. Furthermore, the findings indicate that modern technological evaluation must seamlessly integrate quantitative risk assessments with qualitative impact assessments to ensure both physical safety and ethical fairness. Compliance should not be viewed as a bureaucratic limitation on future innovations, but rather as a critical enabler of digital trust. To ensure the sustainable and safe integration of emerging technologies into society, regulatory compliance, continuous risk management, and ethical frameworks must become foundational components of engineering, medical, and information technology education curricula.

Anna Bychkova

(Moscow Technological Institute)

Cybersecurity Competencies for AI-Enabled Medical Technologies: A Framework for MedTech Education

The increasing integration of artificial intelligence, connected medical devices, cloud services, and data-driven technologies into healthcare is changing the cybersecurity competencies required of MedTech professionals. Traditional approaches focused primarily on information security and data protection are no longer sufficient when medical technologies incorporate AI models whose security, robustness, transparency, and behaviour may directly affect system reliability and patient safety.

This contribution examines the cybersecurity knowledge and skills that should be incorporated into contemporary MedTech education. Drawing on recent literature, regulatory requirements, and emerging practices for AI-enabled medical technologies, it proposes a competency framework connecting cybersecurity with data protection, AI security, and trustworthy AI.

The proposed framework covers several interconnected areas: security-by-design for connected medical technologies; identity, access and network security; protection and governance of sensitive medical data; AI-specific threats such as adversarial manipulation and data poisoning; privacy-preserving approaches; model robustness and explainability; lifecycle monitoring and incident response; and human oversight, accountability, and ethical considerations.

Rather than treating cybersecurity, artificial intelligence, data protection, and ethics as separate educational topics, the framework proposes integrating them through interdisciplinary and scenario-based learning. Learners could examine realistic situations involving connected or AI-enabled medical technologies and evaluate technical vulnerabilities alongside privacy, safety, and ethical consequences.

The framework is intended to support discussion on curriculum development and the preparation of MedTech professionals for increasingly connected and AI-enabled healthcare environments.

Azar Mamiyev

(Obuda University)

*Cost as an Unmonitored Security Signal: Economic Denial of Sustainability Attacks
Against Fixed-Budget Cloud Learning Platforms*

Cloud services that scale on demand are billed by use, so an adversary can inflate an operator's bill without degrading availability. Availability alerts stay silent, because nothing has become unavailable. The attack class is known as Economic Denial of Sustainability, or Denial of Wallet. For a platform funded from a fixed budget rather than an elastic cost, reaching the spending ceiling means the service is throttled or suspended, so financial damage ends in loss of access.

We mapped the requirements this attack imposes onto fourteen SOC-CMM capability dimensions and cross-checked the result against NIST SP 800-53 Revision 5 and CSA Cloud Controls Matrix v4. Eleven dimensions record no coverage of cost-based threats and three record partial coverage, and neither control catalogue treats cost as a security signal. At published pricing a sustained attack accrues roughly 893 USD per day undetected, while operational counters published every minute bound the same attack at roughly 9 USD.

We propose Cloud Financial Security as a cross-cutting capability with a six-level maturity ladder, and outline one practical laboratory session suitable for trainer preparation.

Dr. Eszter Metz

(Sapientia Hungarian University of Transylvania)

Teaching Data and Network Visualization in Public Health Education: A Ten-Module Course for Building Systems Thinking

Public health problems are rarely isolated: they emerge from systems of actors, institutions, behaviours and information flows. Curricula, however, still largely train students to read isolated indicators, leaving graduates under-prepared for relational reasoning and critical visual interpretation. This contribution presents the design and outcomes of a one-semester course that embeds data visualization and network science into an undergraduate public health programme, with the pedagogical aim of shaping systems thinking rather than transmitting software skills.

The course runs within the Bachelor-level Public Health programme of Sapientia Hungarian University of Transylvania (180 ECTS; cohorts of approximately 20 students). It comprises ten modules: data and the attention economy; the data-analysis model; data collection from Romanian and European databases; data cleaning and preparation; principles of data visualization; network science and network visualization; misinformation and design criteria; graphics, colour and perception; maps and spatial epidemiology; and infographics. Each module pairs a short theoretical input with hands-on analysis of real public health datasets. Students construct and interpret network graphs of services and institutions, apply centrality measures to identify key actors, and use clustering to reveal at-risk communities and inequalities in access. A dedicated strand on data deception trains the critical reading of truncated axes, cherry-picking, spurious correlation and false balance, framed by Bertin's criteria of expressive and effective representation. Assessment is portfolio-based: each student produces a visual and network analysis of a self-selected public health question.

Across cohorts, students shift from indicator-based to relational description: they identify structural vulnerabilities and key actors rather than reporting single values, justify their graphical choices, and detect misleading encodings in published material. Network visualizations function as a shared language across professional backgrounds, supporting interdisciplinary dialogue.

Data and network visualization is not a technical add-on but an essential competency for analytical clarity and systems-oriented reasoning. The ten-module structure is transferable, and datasets and exercises can be shared on request.

Dr. Ágnes Sántha

(Sapientia Hungarian University of Transylvania)

Basic descriptive statistical analysis for public health education

Descriptive statistics is fundamental to public health education, providing tools to summarize and interpret population health data. Individual indicators capture different dimensions of population health: health status, disease occurrence and mortality, population groups with greater health needs, as well as changes over time.

Beyond calculation, teaching descriptive statistics should emphasize the critical interpretation of health indicators and their limitations. Data can never be interpreted in isolation. Students are prepared to acknowledge that descriptive associations reveal patterns but do not establish causal relationships. With a hands-on short case study I present how teaching descriptive statistics can guide to the practice of evidence-based public health.

Dr. Fanni Zsarnóczy-Dulházi

(Quant Holding Zrt)

From Houston to the Mátra: Implementing Texas Medical Center Accelerator Experience in a Hungarian MedTech Living Lab and Train-the-Trainer Programme

MedTech education delivers technical, clinical and regulatory content in separate curricular silos, while the competences practitioners actually need — protocol design, MDR-compliant data acquisition and evidence generation under real-world conditions — are acquired almost exclusively in practice. International living labs and accelerator programmes have each solved parts of this problem, but their solutions are rarely translated into an operating training facility in Central Europe.

This contribution reports what has actually been implemented from three international reference environments in the Mátra Home Care Living Lab, established by Quant Holding Zrt. in Gyöngyös in 2025, and how each transferred element now functions as a teaching component within a blended-learning and Train-the-Trainer programme for MedTech education.

Comparative observational model analysis was combined with implementation documentation. Three reference environments were analysed based on direct professional engagement: Dutch premium and dementia-specialised elderly care and the Finnish community home care programme in Pori, both examined jointly by the two authors, and the AI-focused mHealth accelerator of Texas Medical Center Innovation (TMCi, Houston, USA), in which the second author participated. The derived design principles were operationalised in a pilot cohort of N=30 (mean age 64 years; 20 female, 10 male) over five days of continuous monitoring using exclusively MDR- certified instrumentation within the pilot mHealth platform.

Three implementation strands are now operational. From the Dutch model, technology was subordinated to the care relationship: passive, non-wearable sensing was adopted and sensor placement is governed by a care-first protocol, which now forms the technology-acceptance teaching module. From the Pori model, explicit data-quality checkpoints and a five-day acquisition window were implemented for participants with limited digital literacy, and the resulting imperfect datasets serve as authentic teaching material on real-world data quality. From the TMCi accelerator, milestone- based mentoring replaced linear content delivery, an MDR-first device policy was adopted, and validation evidence is framed as a market-readiness signal — together the basis of the regulatory- literacy and evidence-communication modules. The pilot was completed without dropouts, protocol deviations or device failures, with complete cardiovascular and sleep datasets acquired for all thirty participants. Academic validation is supplied by the Department of Kinesiology, Hungarian University of Sports Science, whose expertise in movement analysis and musculoskeletal rehabilitation is embedded directly in the training content.

A privately operated living lab can function simultaneously as MedTech validation infrastructure and as a training venue, provided that MDR-grade protocols, academic partnership and international benchmarking are combined. Implementation continues with the N=50 validation cohort currently in acquisition, three curriculum modules derived from the transfer strands above, and a trainer certification pathway. Quant Holding Zrt. positions the Mátra Home Care Living Lab as a regional Central European MedTech reference site and offers the resulting replication package to consortium partner

Victor Kwabla Senaya

(Obuda University)

Virtual PK-PD digital twins for oncology education: A simulation-based framework for technical, clinical, and regulatory

Medical technology education increasingly requires learners to integrate technical, clinical, and regulatory knowledge, rather than acquiring these domains independently. Digital twins and Pharmacokinetics-Pharmacodynamics (PK-PD) modelling offer a simulation-based means of exposing learners to model-informed treatment decision-making; however, their use as an integrated educational platform for oncology competency development remains underexplored. This creates an opportunity to use virtual PK-PD digital twins as a safe simulation environment in which learners can examine how computational models may support cancer-treatment decision-making while recognizing model uncertainty, limitations, and regulatory considerations. This paper proposes a simulation-based educational framework using virtual PK-PD digital twins of cancer patients to develop three complementary competency domains: Technical competency - understanding PK-PD model construction, parameterization, simulation, and uncertainty analysis; Clinical competency - interpreting simulated treatment response and toxicity within a patient-specific context; Regulatory competency - understanding model credibility, validation requirements, limitations, and the appropriate use of computational models in regulated healthcare and drug-development settings. The proposed framework conceptualizes the virtual PK-PD digital twin as a virtual oncology laboratory within which learners work with simulated patient profiles and PK-PD models. Learning activities include parameterizing patient-specific models, simulating alternative treatment and dosing scenarios, examining predicted drug exposure and treatment response, assessing toxicity and uncertainty, interpreting model outputs from a clinical perspective, and considering regulatory requirements associated with model-informed decision-making. The framework is designed for blended learning, combining self-paced e-learning modules with instructor-guided simulation activities. The learning pathway connects PK-PD modelling and digital-twin simulation with clinical interpretation and regulatory evaluation, thereby integrating technical, clinical, and regulatory learning within a common simulation environment. As this work presents a proposed educational framework rather than a completed controlled educational intervention, quantitative learner-performance results are not yet available. The principal outcome is the conceptual integration of technical, clinical, and regulatory competencies within a single virtual simulation environment. The framework provides a structured setting in which learners can explore alternative treatment scenarios without exposing real patients to experimental dosing decisions. Importantly, uncertainty, model limitations, and the credibility of computational predictions are incorporated into the learning process rather than treated as separate considerations.

The proposed framework positions virtual PK-PD digital twins as educational infrastructure for MedTech education, extending their potential role beyond research and clinical decision-support applications. By integrating computational modelling, clinical interpretation, and regulatory considerations within a simulation-based learning environment, the framework addresses the interdisciplinary competencies required for emerging model-informed healthcare practice. Its principal contribution is a technical-clinical-regulatory educational framework that can help learners understand not only how computational oncology models are constructed and simulated, but also how their outputs should be interpreted, how uncertainty and limitations should be recognized, and why regulatory and credibility considerations are essential to their responsible application.

Prof. Dr. László Vasa

(Széchenyi István University);

New World, New Challenges

Artificial intelligence is rapidly transforming not only how knowledge is accessed, but also how people learn, make decisions, cooperate, govern and provide professional services. The central challenge of this transformation is therefore not simply technological adaptation, but the preservation and strengthening of human agency. As AI systems evolve from information-retrieval tools toward generative and increasingly agentic technologies, education must equip learners and professionals not only to use these systems effectively, but also to question their outputs, understand their limitations and retain responsibility for consequential decisions.

This plenary presentation examines the implications of the AI age for higher education, professional training and MedTech education in particular. It argues that human judgment, responsibility and oversight must remain fundamental principles of responsible AI use. In education, the growing presence of AI is transforming the traditional relationship between teacher and student into a new learning environment in which AI may assist, personalize and simulate, while educators remain responsible for contextualisation, assessment, mentoring and the development of critical competence. Particular attention is given to the tension between the opportunities and risks of AI-supported learning. While AI can enable personalised learning pathways, simulation-based practice, more efficient use of faculty time and the development of AI fluency, excessive dependence on easily generated outputs may undermine active thinking, academic integrity and the deeper learning processes through which durable competence is formed. In MedTech education, these questions carry heightened significance because errors and poor judgment may affect patients, professionals, institutions and public trust. Train-the-Trainer models must therefore integrate data literacy and technical competence with clinical judgment, regulatory awareness, ethical reasoning and human trust.

The presentation also places educational transformation within a broader discussion of sustainability and technological sovereignty. AI depends on material, institutional and human foundations, including energy, critical resources, infrastructure, skills and resilient institutions. Universities consequently have a strategic role extending beyond knowledge transmission: they must develop research capacity, digital competence, ethical judgment, cooperation and resilience.

The central message is that AI should enhance rather than displace human capability. The future of education will depend on our ability to combine technological innovation with responsibility, trust, human connection and the cultivation of qualities that machines cannot replace.

Prof. Dr. Borsos Éva

(University of Novi Sad, Hungarian Language Teacher Training Faculty, Subotica)

A tantermen kívüli oktatás a mesterséges intelligencia korában

A tantermen kívüli oktatás – bármelyik meghatározást is vesszük alapul - fő célja a természet közvetlen, érzékszervek által történő megismerése. A mesterséges intelligencia (MI) oktatásban megvalósuló rohamos térnyerése a tantermen kívüli oktatást is elérte. Nagyon fontos, hogy a mesterséges intelligencia még véletlenül se vegye át a tantermen kívüli oktatás helyét vagy szerepét az iskolai nevelésben, hanem csak és kizárólag módszertanilag támogassa.

A munka azt vizsgálja, miként őrizhető meg a tantermen kívüli oktatás lényege a tapasztalatokon keresztül való ismeretszerzés egy olyan technológiai korban, ahol az MI-alapú rendszerek alkalmazása a környezeti nevelésben is mindennapossá vált. A kutatás abból a feltételezésből indul ki, hogy a tantermen kívüli oktatás lényege a természetben való aktív jelenlét, a technológiának pedig kizárólag az előkészítő és a végső kielemező fázisban lehet szerepe, megakadályozva, hogy a diákok képernyőfüggősége a szabadban, a természetben is érvényre jusson.

A tanulmány részletesen bemutatja az MI segítő szerepét a természetközeli tanulás tervezési és értékelési fázisaiban. A pedagógusok intelligens algoritmusok segítségével a helyszíni ökoszisztémára szabott, egyéni differenciált feladatsorokat és megfigyelési szempontokat tervezhetnek meg a kirándulások előtt, minimalizálva a terepen szükséges digitális eszközhasználatot. A diákok a természetben a hagyományos módszerekkel – gyűjtéssel, méréssel, észleléssel – dolgoznak, míg a mesterséges intelligencia a tanterembe visszatérve segíti a gyűjtött adatok, minták komplex elemzését és vizualizációját. Az MI így nem elvonja a figyelmet a növény- és állatvilágról, hanem elmélyíti a terepen szerzett közvetlen tapasztalatok feldolgozását.

A kutatás konkrét példán keresztül mutatja be a mesterséges intelligencia alkalmazását a tantermen kívüli terepi munka során. Külön kiemelve, hogy ne vigyük túlzásba a mesterséges intelligencia használatát, mert, ha ezek az alkalmazások uralják a terepmunkát, az meggátolja a valódi, élményalapú kompetenciafejlesztést, illetve a tapasztalatszerzésre épített ismeretszerzést.

Összességében a tantermen kívüli oktatásnak a mesterséges intelligencia korában sem szabad veszítenie a lényegét: a természetközpontú karakterét. A jövő környezetpedagógiájának legnagyobb kihívása olyan hibrid modellek alkalmazása, ahol a mesterséges intelligencia megmarad hatékony módszertani háttéreszköznek, miközben a tanulás valódi színtere és fókusza maga a természet marad, a benne élő növényekkel és állatokkal együtt.

Pál Fehér-Polgár

(Obuda University)

E-kereskedelmi technológiák és innovációs megoldások; AI-alapú személyre szabás és interaktív e-learning módszertanok használata a felsőoktatásban

A digitális transzformáció és a 2026-ra várható technológiai paradigmaváltások alapjaiban formálják át az e-kereskedelmi platformokat. Az olyan áttörést jelentő technológiák, mint a generatív mesterséges intelligencia (AI), a kiterjesztett és virtuális valóság (AR/VR), a biometrikus rendszerek, valamint az intelligens ellátási láncok nemcsak a lakossági és B2B értékesítést forradalmasítják, hanem teljesen új kihívások és lehetőségek elé állítják a felsőoktatást és a szakképzést is.

Jelen előadás azt vizsgálja, hogy a közelmúlt e-kereskedelmi és digitális innovációi hogyan integrálhatók oktatási tananyagokba és e-learning / blended learning struktúrákba. Az előadás kiemelten foglalkozik az alábbi területekkel: 1. AI-alapú adaptív tanulási utak és interaktív asszisztensek: Hogyan alkalmazhatók az e-kereskedelemben bevált hyper-perszonalizációs algoritmusok a hallgatók és képzők egyéni tanulási igényeinek támogatására. 2. AR/VR és 3D-s termékszimulációk a digitális pedagógiában: A virtualizált termékbemutatók és “try-before-you-buy” technológiák oktatási célú bemutatása hybrid/blended környezetben. 3. Train-the-Trainer és oktatói kompetenciák: Milyen új digitális és technológiai kompetenciákkal kell rendelkezniük az oktatóknak ahhoz, hogy a gyorsan változó piaci és technológiai trendeket zökkenőmentesen átültethessék a tananyagfejlesztésbe és a számonkérési rendszerekbe.

Az előadás gyakorlati esettanulmányokon és jógyakorlatokon keresztül mutatja be, hogy a technológiai innovációk és a digitális pedagógia ötvözése miként segíti a felsőoktatási tudásátadást.

Dr. Szikora Péter

(Obuda University)

A kattintásszámokon túl: az elméleti témakörök lefedettsége mint tanulási stratégia indikátora az informatikai oktatásban

A tanulmány azt vizsgálja, hogy az elméleti témakörök lefedettsége alkalmas lehet-e a hallgatók tanulási stratégiájának jellemzésére egy Moodle-alapú, az adatbázis-tervezésre kiemelt hangsúlyt helyező informatikai kurzuson. A tanulási analitika gyakran olyan mutatókra támaszkodik, mint a kattintások száma, a hozzáférések gyakorisága vagy a tesztkitöltési kísérletek száma. Ezek azonban nem feltétlenül mutatják meg, hogy a hallgatók az egyes témakörökhöz rendelt elméleti tananyagforrásokat milyen mértékben dolgozzák fel.

A kutatás egy 2026-ban megtartott egyetemi kurzus teljes tanulási és értékelési adatállományára épül ($N = 122$). Az elemzés során olyan témakör-lefedettségi mutatókat alakítottunk ki, amelyek azt fejezik ki, hogy a hallgatók az egyes témákhoz tartozó elméleti források mekkora hányadához fértek hozzá. A hallgatókat a lefedettség mértéke alapján passzív, részleges és teljes lefedettségű csoportokba soroltuk, majd teljesítményüket Spearman-féle rangkorrelációval, Kruskal–Wallis-próbával és páronkénti Mann–Whitney-összehasonlításokkal elemeztük.

Az eredmények azt mutatják, hogy az elméleti témakörök nagyobb mértékű lefedettsége pozitív kapcsolatban áll mind a tesztalapú, mind az írásbeli számonkérések eredményeivel, míg a teljesen kihagyott témakörök száma negatív összefüggést mutat a teljesítménnyel. A csoportok összehasonlítása szintén arra utal, hogy a magasabb lefedettségi kategóriákhoz kedvezőbb teljesítménymintázatok társulnak, különösen a tesztalapú számonkérések esetében.

A vizsgálat alapján a témakörök lefedettsége a Moodle-ban rögzített tanulási aktivitás strukturáltabb értelmezését teszi lehetővé, mint az egyszerű hozzáférési vagy kattintásszámok. Az informatikai oktatásban, különösen az adatbázisokkal kapcsolatos kurzusokon, az ilyen mutatók elősegíthetik a töredezett vagy hiányos tanulási mintázatok korai felismerését, és támogatják az oktatókat annak nyomon követésében, hogy a hallgatók milyen széles körben dolgozzák fel az előírt elméleti tananyagot.

Dr. Keszthelyi András

(Obuda University)

Korai teljesítményjelzések egy informatikai kurzuson: kurzusszintű eredmények a veszélyeztetett hallgatók azonosítására

A tanulmány azt vizsgálja, hogy egy gazdaságinformatikus képzés informatikai kurzusán a félév korai szakaszában rendelkezésre álló teljesítménymutatók milyen módon segíthetik a részvétel csökkenése, illetve a kurzus sikertelen teljesítése szempontjából veszélyeztetett hallgatók azonosítását. Az elemzés a 2025. őszi félévben a kurzust felvevő 144 hallgató teljesítményadataira épül. A vizsgálat a heti feladatok eredményeit, az aktivitási mutatókat, a félévközi számonkérésen való részvételt és a végső érdemjegy megszerzését veszi figyelembe annak feltárására, hogy a félév elején teljesített feladatok eredményei használhatók-e az oktatók számára gyakorlati korai figyelmeztető jelzésként.

Az eredmények alapján a kezdeti feladatokon nyújtott gyengébb teljesítmény alacsonyabb későbbi aktivitással, a kurzusfeladatokból való korábbi lemorzsolódással, a számonkéréseken való kisebb részvétellel, valamint a végső érdemjegy megszerzésének alacsonyabb valószínűségével jár együtt. Ezzel párhuzamosan azok a hallgatók, akik a félév során tartósan magasabb aktivitást mutattak, kedvezőbb eredményeket értek el mind a félévközi, mind a végső értékelési mutatók tekintetében.

A vizsgálat eredményei arra utalnak, hogy az egyszerű, kurzusszinten rendelkezésre álló adatok – különösen a korai feladatok eredményei és az aktív hetek száma – alkalmasak lehetnek azon hallgatók időben történő azonosításának támogatására, akiknek további oktatói segítségre lehet szükségük. A kutatás a gyakorlatorientált tanulási analitika területéhez azzal járul hozzá, hogy bemutatja: a kurzusok során rutinszerűen keletkező adatok nemcsak az utólagos értékelésben, hanem a korai beavatkozás és a hallgatói támogatás megalapozásában is felhasználhatók.

Ildikó Kertai-Kiss

(Károli Gáspár University)

A reflektív kompetenciák a személyes tanulásban

A reflexió, mint kulcskompetencia társadalmi hasznossága, alkalmazhatósága elengedhetetlen a bizonytalanság és az AI korában. Az élethosszig tartó személyes tanulás és a változás fontos eleme, hiszen nem egyszerűen a tapasztalatokból tanulunk, hanem a reflektált tapasztalatokból (Dewey, 1951). Az OECD felmérése alapján a szociális és érzelmi készségek iránti kereslet 2030-ra 11-14%-kal emelkedik, az új munkahelyek vonatkozásában pedig globálisan 2030-ig az egészségügy, szociális munka, mentálhigiéné vezetik a növekedést. (WEF Future of Jobs, 2025) Ezekben az ágazatokban, mind a szakemberek képzésében, mind a segítő hivatás gyakorlásához a reflektivitás mindennapos gyakorlat, folyamatosan fejlesztendő kompetencia. Az előadás első része ismerteti a reflektív kompetencia mibenlétét, szerepét, melynek lényege, hogy interakciót hozzon létre a reflektáló személy és a problémás szituáció között, olyan szándékos, célirányos, strukturált gondolkodás, melynek célja a fejlődés és a változás. Tehát a reflexió tudásunk és vélekedésünk aktív, kitartó és pontos vizsgálata és a belőlük származó következtetések megfontolása és jövőbeni alkalmazása a gyakorlatban. Fontos hozadéka, hogy támogatja a kritikus gondolkodást, az adaptivitást és a rezilienciát, segíti a problémamegoldást, fejleszti az EQ-t és a kreativitást, valamint a kapcsolódást önmagunkhoz, egyedi látásmódunkhoz, kognitív és érzelmi erőforrásainkhoz. Összességében tehát egy olyan felnőttkori tanulási modell, transzformáció, melynek eredménye egy új szemlélet és a rugalmas perspektíva váltás képessége. Az előadás második része kitekintést ad a reflektivitás és az AI kapcsolatára és azokat a nem automatizálható emberi készségeket vizsgálja, amelyekre a mesterséges intelligencia nem képes. Ezek között szerepel a kapcsolati hangolódás (Attunement), mely a tükröneuronok működésén alapul és többek között azért lényeges, mert a munkaadók 71%-a fontosabbnak tartja az EQ-t, mint a technikai tudást, az etikai ítélőképesség (az AI szabályokat követ, de nem bonyolult erkölcsi dilemmákban dönt), tehát a jövő szakembere az „etikai iránytű” szerepét tölti be, a konténerelés (Containing), más szóval az intenzív, zavaros érzelmi tartalmak befogadása és megtagadása (az AI ambivalenciát nem visel el, de a professzionális segítő szalmákban dolgozók igen), a jelenlét, mint kompetencia (neurobiológiai önszabályozás, szomatikus rezonancia képessége). Ezek a tudások fejleszthetők, mérhetők és 2030-ra az alkalmazott pszichológia területein a képzés kötelező részei lesznek. Végül szó esik az AI megjelenéséről napjainkban a pszichopiacon, egyrészt az oktatásban, képzésben (pl. kérdezéstechnika tanítása), emellett a napi gyakorlati munkában már most érzékelhető, hogy az AI-jal kapcsolatos mentális zavarok kiemelt terület lesz.